

Classical Programming Languages: The Story of Fortran



Recent research to prepare a list of the top 10 popular programming languages for the year 2021 threw up quite a number of surprises for the author. One was finding out the fact that three programming languages that were thought to be of a bygone era – and hence having historic significance only - are still surviving and making an impact in the programming world. These are: Fortran, COBOL and Pascal. The first article in this three part series discusses Fortran, a programming language still widely used for scientific computing.

The three languages Fortran, COBOL and Pascal are still used professionally, even after being in service for more than half a century. As these programming languages have made quite an impact in the programming world, and have become long-term trend-setters and bearers of standards, I call them classical programming languages.

The first serious question regarding Fortran is: Why is it still relevant? Fortran, which came into existence in 1957, is still ranked 25th and 30th in two of the popular programming language ranking schemes. This makes it better ranked than many programming languages that have captured the public eye recently. Though Fortran was used as a general-purpose programming language in the earlier days, nowadays it can be considered a domain-specific programming language. But what are the reasons behind its success for more than six decades in the field of scientific computing? Of

course, there are many specialised tools and programming languages for scientific computing — MATLAB, Scilab, Maple, Mathematica, Julia, etc, are a few examples. But Fortran has a clear advantage over these tools. A Fortran implementation of a solution to a problem is much faster than a similar implementation in any of the tools mentioned earlier. To illustrate this point further, let us consider the programming languages used to write numerical libraries of Scilab. Of course, one of the programming languages is Fortran. Also, the memory consumed by a Fortran solution is much less when compared with similar solutions in many other popular tools. Certainly, using the tools mentioned earlier, in general, is easier. Still, the faster execution in Fortran is critical, especially when the task is complex in nature. Hence, Fortran coding is still used in disciplines such as physics, especially in large scale simulation of physical systems. This includes astrophysical modelling, computational fluid dynamics, climate modelling, etc.